Chart Chooser Cheat Sheet

Module 8 · Visualization & Storytelling · ShopKart examples use `shopkart_clean.csv`

Step 1 – What is the chart's job?

Job	Question	Use	Avoid	ShopKart example
Compare	Which is bigger?	Bar chart (sorted); horizontal bars for long names	3D bars, pies with many slices	Revenue by category: Electronics ₹73.3 L
Trend	How did it change over time?	Line chart, time on the x-axis	Bars for 12+ time points, unsorted months	Monthly revenue: peak Oct ₹19.8 L (Nov ₹19.7 L), low May ₹7.9 L
Distribution	How are values spread?	Histogram, box plot	Plotting the mean alone	Order revenue: median ₹1,525, mean ₹2,941 (right-skewed)
Compare spreads	Do groups differ?	Box plot by group	Two separate histograms with different axes	Delivery days: median 5 (returned) vs 4 (kept)
Relationship	Do two numbers move together?	Scatter plot (alpha, jitter)	Line chart for unrelated points	Discount vs quantity, r = +0.27
Many relationships	Which pairs move together?	Correlation heatmap (−1 to +1)	Rainbow colour scale	delivery_days vs rating, r = −0.33
Part of whole	What share is each part?	Bar of %; pie or donut only for 2–3 parts	3D/exploded pies, 10-slice pies	City share: Delhi 15.8%
One number	How are we doing?	KPI card (big number + label)	A chart for a single value	Return rate 13.7%

Step 2 – Clean-chart rules

1. Bars start at **zero**.
2. **Sort** bars, unless the order has meaning (months, age bands).
3. Label both axes **with units**: ₹ lakh, %, days.
4. **One highlight colour**; everything else grey or one base colour.
5. Same colour = same meaning across all charts (e.g. orange = returned).
6. No 3D, no shadows; light gridlines or none.
7. **Message title** = the finding as a sentence with a number.
8. Five-second test: can someone get the point in 5 seconds?

Step 3 – Code

Matplotlib

```
import matplotlib.pyplot as plt
s = df.groupby("category")
["revenue"].sum().sort_values(ascending=False) / 100000
plt.figure(figsize=(8, 4))
plt.bar(s.index, s.values)                    # plt.barh for horizontal
plt.title("Electronics brings half of revenue")
plt.xlabel("Category"); plt.ylabel("Revenue (₹ lakh)")
plt.tight_layout()
plt.savefig("images/chart.png", dpi=150)    # BEFORE plt.show()
plt.show()
```

Chart	Call
Line	`plt.plot(x, y, marker="o")`; then `plt.ylim(0, ...)`, `plt.xticks(pos, labels)`
Histogram	`plt.hist(df["revenue"], bins=40)`
Reference line	`plt.axhline(value, linestyle="--")` or `plt.axvline(...)`
Side by side	`fig, axes = plt.subplots(1, 2, figsize=(12, 4))`, then `axes[0].set_title(...)`
Rotate labels	`plt.xticks(rotation=45)`

Seaborn

```
import seaborn as sns
sns.countplot(data=df, x="category", hue="returned")        # counts rows
sns.barplot(data=df, x="category", y="revenue")             # MEAN + confidence interval
sns.boxplot(data=df, x="returned", y="delivery_days")
sns.scatterplot(data=df, x="discount", y="quantity", hue="membership", alpha=0.4)
sns.heatmap(df[num_cols].corr(), annot=True, cmap="RdBu_r", vmin=-1, vmax=1)
```

Power BI quick reference

Task	How
Load CSV	Home → Get data → Text/CSV → Transform data (check types) → Close & Apply
Measure	Right-click table → New measure
Total Revenue	`SUM(shopkart_clean[revenue])` = ₹1,47,04,649.70
Total Orders	`COUNTROWS(shopkart_clean)` = 5,000
Return Rate %	`DIVIDE(CALCULATE(COUNTROWS(shopkart_c`

Task	How
	`lean), shopkart_clean[returned] = "Y"), COUNTROWS(shopkart_clean))` = 13.74%
Avg Rating	`AVERAGE(shopkart_clean[rating])` = 3.96 (ignores blanks)
Total Profit	`SUM(shopkart_clean[profit])` = ₹31,40,749.70
Slicer	Visualizations → Slicer → drag field
Export	File → Export → Export to PDF

Mac users: Tableau Public (free). Same ideas; use calculated fields instead of measures.

Storytelling template

Part	Question	ShopKart example
Context	What does the audience already know?	Return rate is 13.7% overall
Insight	What is the one surprising thing?	Pune returns are 2.4x the average (32.4%, 146 of 451 orders)
Evidence	Which chart and numbers prove it? What did you check?	Higher at every delivery speed (2–4 days: 31.7% vs 10.5%)
Recommendation	What should we do, and who does it?	Audit Pune Fashion (38.8% returned); pilot a faster courier as an A/B test

Topic title → message title

- "Monthly Revenue" → "Revenue peaks in October at ₹19.8 lakh"
- "Revenue by Category" → "Electronics brings half of all revenue"

Correlation is a clue, not proof: recommend a test.